Landmark Heart Failure Trials – Evidence Summary

This document summarizes the major randomized controlled trials that established modern guideline■directed medical therapy (GDMT) for heart failure, including ACE inhibitors, ARBs, ARNIs, beta■blockers, mineralocorticoid receptor antagonists (MRAs), and SGLT2 inhibitors. Trials are presented with population, intervention, and key outcomes.

1. ACE Inhibitor Trials

Trial	Population	Drug / Dose	Key Outcomes
CONSENSUS (1987)	NYHA IV HFrEF	Enalapril	↓ Mortality ~40% at 6 months
SOLVD■Treatment (1991)	Chronic HFrEF	Enalapril	↓ Mortality, ↓ HF hospitalizations
SOLVD■Prevention (1992)	Asymptomatic LV dysfunction	Enalapril	↓ Progression to HF
SAVE (1992)	Post■MI, EF ≤40%	Captopril	↓ CV death, ↓ HF
AIRE (1993)	Post■MI with HF	Ramipril	↓ Mortality
TRACE (1995)	Post■MI, EF ≤35%	Trandolapril	↓ All■cause mortality
ATLAS (1999)	Chronic HFrEF	High vs low■dose lisinopril	↓ HF hospitalizations; dose matters

2. ARB Trials

Trial	Population	Comparison	Key Outcomes
ELITE II (2000)	Chronic HFrEF	Losartan vs captopril	ARB not superior; better tolerated
CHARM■Alternative	HFrEF, ACEi■intolerant	Candesartan vs placebo	↓ CV death/HF hospitalization
CHARM■Added	HFrEF on ACEi	Add candesartan	↓ HF events; ↑ adverse effects
VAL■HeFT	Chronic HFrEF	Valsartan vs placebo	↓ HF hospitalizations
VALIANT (2003)	Post■MI + HF/LV dysfunction	Valsartan vs captopril	ARB non■inferior; combo harmful
ONTARGET (2008)	High CV risk (not HF)	Telmisartan vs ramipril	ARB non■inferior; combo harmful

3. ARNI (Sacubitril/Valsartan) Trials

Trial	Population	Comparison	Key Outcomes
PARADIGM■HF (2014)	Chronic HFrEF	ARNI vs enalapril	↓ CV death & HF hospitalization
PIONEER■HF	Hospitalized HFrEF	ARNI vs enalapril	Rapid NT■proBNP reduction
TRANSITION	Post■acute HF	Early vs delayed ARNI	Early initiation safe
PARAGON■HF	HFpEF	ARNI vs valsartan	↓ HF hospitalizations (subgroups)
PARADISE■MI	Post■MI + LV dysfunction	ARNI vs ramipril	No clear superiority

4. Beta■Blocker Trials

Trial	Drug	Population	Key Outcomes
MERIT■HF	Metoprolol succinate	HFrEF	↓ Mortality ~34%
CIBIS■II	Bisoprolol	HFrEF	↓ Mortality ~34%
COPERNICUS	Carvedilol	Severe HFrEF	↓ Mortality ~35%
US Carvedilol	Carvedilol	HFrEF	Major mortality reduction
COMET	Carvedilol vs metoprolol tartrate	HFrEF	Carvedilol favored*
SENIORS	Nebivolol	Elderly HF	↓ Death/HF hospitalization

5. Mineralocorticoid Receptor Antagonist Trials

Trial	Drug	Population	Key Outcomes
RALES	Spironolactone	Severe HFrEF	↓ Mortality 30%
EPHESUS	Eplerenone	Post■MI + HF/LV dysfunction	↓ Mortality, ↓ SCD
EMPHASIS■HF	Eplerenone	Mild HFrEF (NYHA II)	↓ Death/HF hospitalization
TOPCAT	Spironolactone	HFpEF	↓ HF hospitalizations

6. SGLT2 Inhibitor Trials

Trial	Drug	EF Range	Key Outcomes
DAPA■HF	Dapagliflozin	≤40%	↓ CV death/HF hospitalization
EMPEROR■Reduced	Empagliflozin	≤40%	↓ HF hospitalization; renal protection
EMPEROR■Preserved	Empagliflozin	>40%	↓ HF hospitalization
DELIVER	Dapagliflozin	>40%	↓ HF events across EF spectrum
EMPULSE	Empagliflozin	Any EF (acute HF)	Early clinical benefit
SOLOIST■WHF	Sotagliflozin	Any EF (diabetic HF)	↓ CV death/HF hospitalization

Key principle: Modern heart failure therapy is built on multiple complementary pillars proven by randomized trials. Outcomes improve when ACEi/ARB/ARNI, beta■blockers, MRAs, and SGLT2 inhibitors are combined appropriately rather than used in isolation.